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Permian and Triassic paleomagnetism of the southwestern Tien Shan: timing and mode of tectonic rotations

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ABSTRACT

Volcanic rocks of the Lower Permian Luchob Formation and sedimentary rocks of the presumed Upper Permian Hanaka and Middle–Upper Triassic Madighen formations were studied in three localities in the southwestern Tien Shan. Primary magnetizations were isolated in Lower Permian and Triassic rocks, whereas the remanence of reversed polarity in the Hanaka Formation acquired during the Kiaman superchron may be slightly younger than the rock age. Inclinations in all three formations agree with the European reference data, but declinations are deflected westward. We argue that these deflections are due to the Late Permian–Late Triassic (but pre-Rhaetian) counterclockwise rotations and partly due to Late Cenozoic counterclockwise rotations. Inclinations in Permian and Lower Triassic rocks throughout the Tien Shan foldbelt, the Junggar basin and north Tarim are also in agreement with the reference data, implying that, within the error limits, this part of Central Asia was already welded together with Kazakhstan and the European plate in the Permian. Westward-deflected declinations throughout this region are accounted for not by movements of large blocks such as the Tarim but by domino-fashion Late Permian–Triassic rotations by strike-slip.

1. Introduction

The most reliable tectonic interpretation of paleomagnetic data is achieved if there are results from well-dated formations covering a long time span. In such cases we can both estimate magnitudes of tectonic movements and date them. To fill long time gaps in the paleomagnetic record it may sometimes be necessary to study poorly dated formations. The importance of data in such gaps may be so great that any opportunity to find paleomagnetic 'missing links' should be used.

The Tien Shan foldbelt in Central Asia underwent several epochs of deformation during the Phanerozoic, and many aspects of its tectonic evolution are controversial. Paleomagnetic data on Cenozoic [1] and Cretaceous [2] rocks from its western part have helped to outline the main features of the Alpine movements in this region, which are most probably connected with the India–Eurasia collision [3]. Paleomagnetic data on Paleozoic rocks revealed rotations of a much larger magnitude [4] but, so far, the age of these movements could only be constrained as post-Early Permian and pre-Jurassic, mainly on geological grounds.

The southwestern part of the Tien Shan lacks reliable paleomagnetic data. Late Paleozoic rocks have been sampled in the the Ghissar range (Fig. 1a) from eighteen sections, but, with the exception of one section, only *in-situ* mean directions have been reported [5]. These means are very scattered, and their regional significance is doubtful. There are some presumed Upper Permian and Triassic rocks in the southern Tien Shan, but the only result on these Triassic sedimentary rocks was presented in an abstract, without neither mean direction nor details of the laboratory treatments [6].

As a part of our paleomagnetic and structural studies in the Tien Shan we sampled pre-Jurassic rocks from different parts of the belt. Below are presented some results on the Permo-Triassic formations from the western part of the south

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Tien Shan which may help to fill the gaps in both the spatial and temporal distribution of the paleomagnetic data.

2. Geological setting and sampling

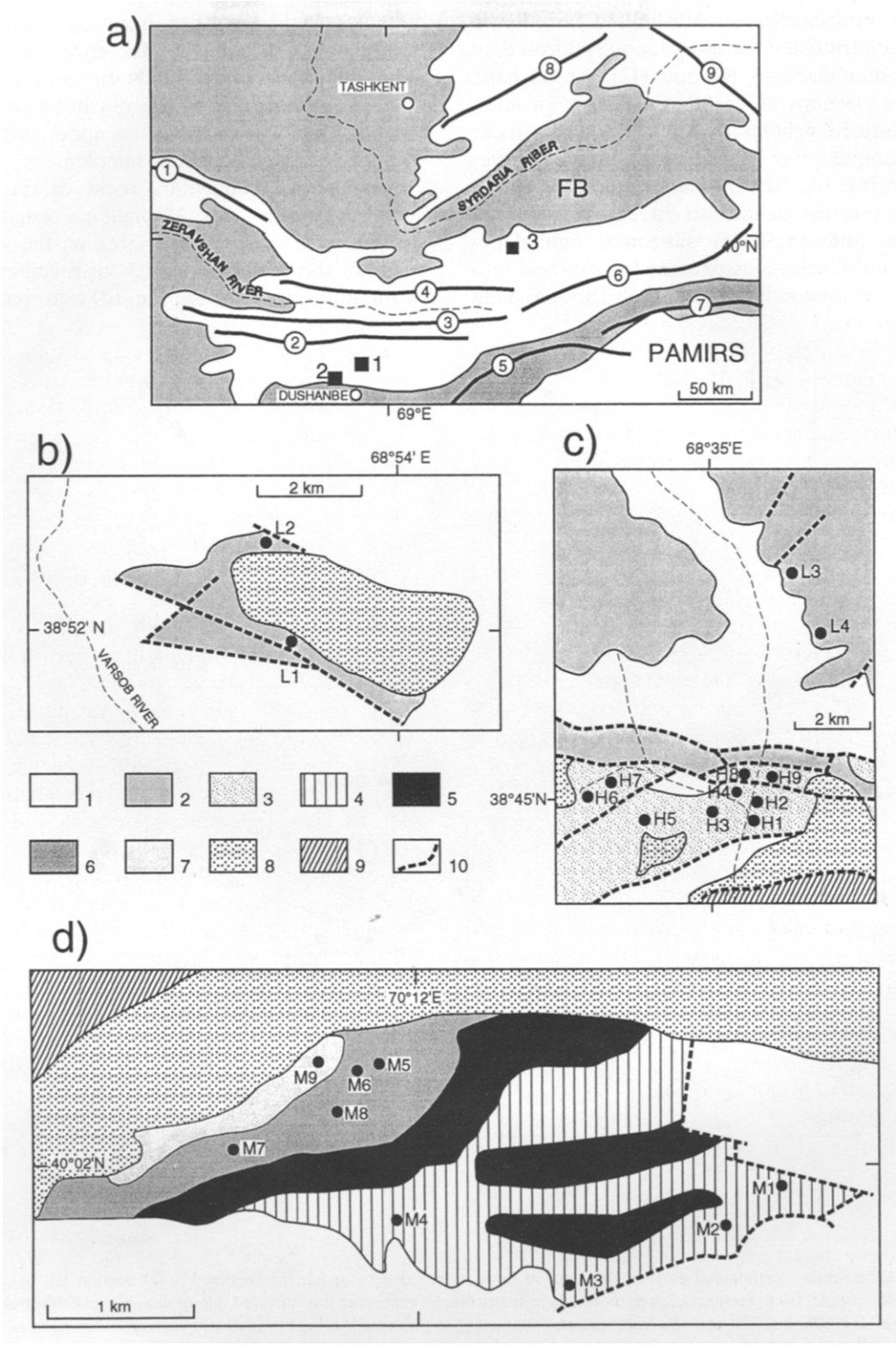
The Tien Shan realm includes Early Paleozoic foldbelts in the north and Late Paleozoic foldbelts in the south. (In the Russian literature these are referred to as Caledonian and Variscan respectively.) It is generally believed that, from the Early Permian, and perhaps form the the beginning of the Late Permian, the tectonic activity almost ceased, and that after this time the foldbelt became nearly passive [e.g., 7]. After the Late Paleozoic activity, a long gap in sedimentation from the Late Permian to the Early Jurassic was common to the entire Tien Shan. Some local formations are believed to have been deposited during this period, but their ages are not well constrained because their volcanic parts are not well dated radiometrically and their redbed components have been dated using only indirect evidence and general correlations [8]. With a major angular unconformity Late Paleozoic and Triassic(?) rocks are overlain by a Jurassic to Paleogene sedimentary sequence which was deformed in the late Cenozoic. An uppermost Triassic (Rhaetian) age has locally been assigned to the basal member of the sequence. Nevertheless, the importance of obtaining Late Paleozoic and Triassic paleomagnetic data is so great that we decided to study these intervals in spite of the imprecise dating. Below are described several formations of presumed Permian and Triassic age from the south Tien Shan which were studied to fill the gaps in the Tien Shan paleomagnetic record.

The Luchob Formation lies with an angular unconformity on mid-Upper Carboniferous rocks and Permian granites in the south Ghissar and is in turn unconformably overlain by Jurassic or Cretaceous sedimentary rocks. The Luchob is about 1000 m thick and consists of andesites, rhyolites, and ignimbrites with some tuffaceous and conglomerate layers. Paleoflora collected from sedimentary beds [8] and K-Ar whole-rock data [9] point to an Early Permian age for these rocks. The volcanic rocks were studied at two localities, and at each locality two long sections were studied (Fig. 1b and c). The rock was very hard, and it was impossible to use the portable drill. Hence only hand samples were taken (53 in total). Most importantly, the volcanic rocks are poorly stratified at the sampled outcrops and although their general structural position was clear the bedding measurements at the sampling points were rather imprecise.

The Hanaka Formation, which has a visible thickness of about 1300 m, is known from a limited area on the southern slope of the Ghissar range (Fig. 1c). No stratigraphic contacts of this formation with older rocks were found but it is certainly younger than the Luchob Formation as well-rounded boulders and pebbles of Luchob volcanic rockss are abundant in the Hanaka conglomerates at all stratigraphic levels. The Hanaka Formation is overlain with an angular unconformity by Jurassic sedimentary rocks; the basal member of these overlying rocks may be Rhaetian in age [8]. The age of the Hanaka is controversial: Some believe that it is as young as Middle–Late Triassic (but pre-Rhaetian) [10], whereas others insist on its Early Permian age [8].

Study of the Hanaka Formation was carried out with two aims in mind: firstly, we wished to obtain a mean direction for the tectonic interpretation, and secondly, the controversy concerning the formation age justified more precise dating, as the Kiaman reversed interval lasted almost throughout the entire Permian, with mixed polarities prevailing subsequently. The main part of the formation consists of violet–brown conglomerates, gritstones and sandstones, generally dip-

Fig. 1. Location map (a) and schematic geological maps of studied localities (b–d). (a) Shaded are the Mesozoic–Cenozoic cover (*FB* = Fergana basin), white are Paleozoic rocks. Main ranges (numbers in circles) are shown as heavy lines: *1* = Nuratau; *2* = Ghissar range; *3* = Zeravshan range; *4* = Turkestan range; *5* = Peter the First range; *6* = Alay range; *7* = Transalay range; *8* = Chatkal range; *9* = Fergana range. (b), (c) and (d) are maps of localities 1, 2 and 3 respectively: *1* = pre-Permian rocks; *2* = Luchob Formation; *3* = Hanaka Formation; *4–7* = Madighen Formation (T1, T2, T3 and T4 members respectively); *8* = Jurassic and Cretaceous rocks; *9* = Paleogene rocks; *10* = faults of all types. Dots are sampled sections.


a)
TASHKENT
SYRDARIA RIBER
FB
ZERAVSHAN RIVER
PAMIRS
DUSHANBE
40°N
69°E
50 km
b)
68°54' E
2 km
38°52' N
VARSOB RIVER
L1
L2
c)
68°35'E
L3
L4
2 km
38°45'N
H1
H2
H3
H4
H5
H6
H7
H8
H9
d)
70°12'E
40°02'N
M1
M2
M3
M4
M5
M6
M7
M8
M9
1 km

ping to north-northwest with limited variation in bedding attitude. These rocks were drilled at twelve sites (sections H2 and H3) and 43 hand samples (sections H3–H6) were taken. In addition, thirteen pebbles of Luchob volcanic rocks were sampled from the Hanaka conglomerates (section H7, Fig. 1c). The upper member of the Hanaka Formation consists of brown–red fine-grained sandstones and siltstones, light limestones and black mudstones and is exposed in a narrow fault-bounded band. Only fifteen hand samples could be taken from this member because of the friability of the rock (sections H8 and H9). The sampled beds dip to the north at the H8 section and to the northeast at the H9 section (Fig. 1c). In total, the upper three quarters of the formation were sampled.

Lacustrine sedimentary rocks of the several hundred metre thick Madighen Formation are known from a very limited area on the southern edge of the Fergana basin. Four members (from bottom to top, T1 to T4; Fig. 1d) were recognized

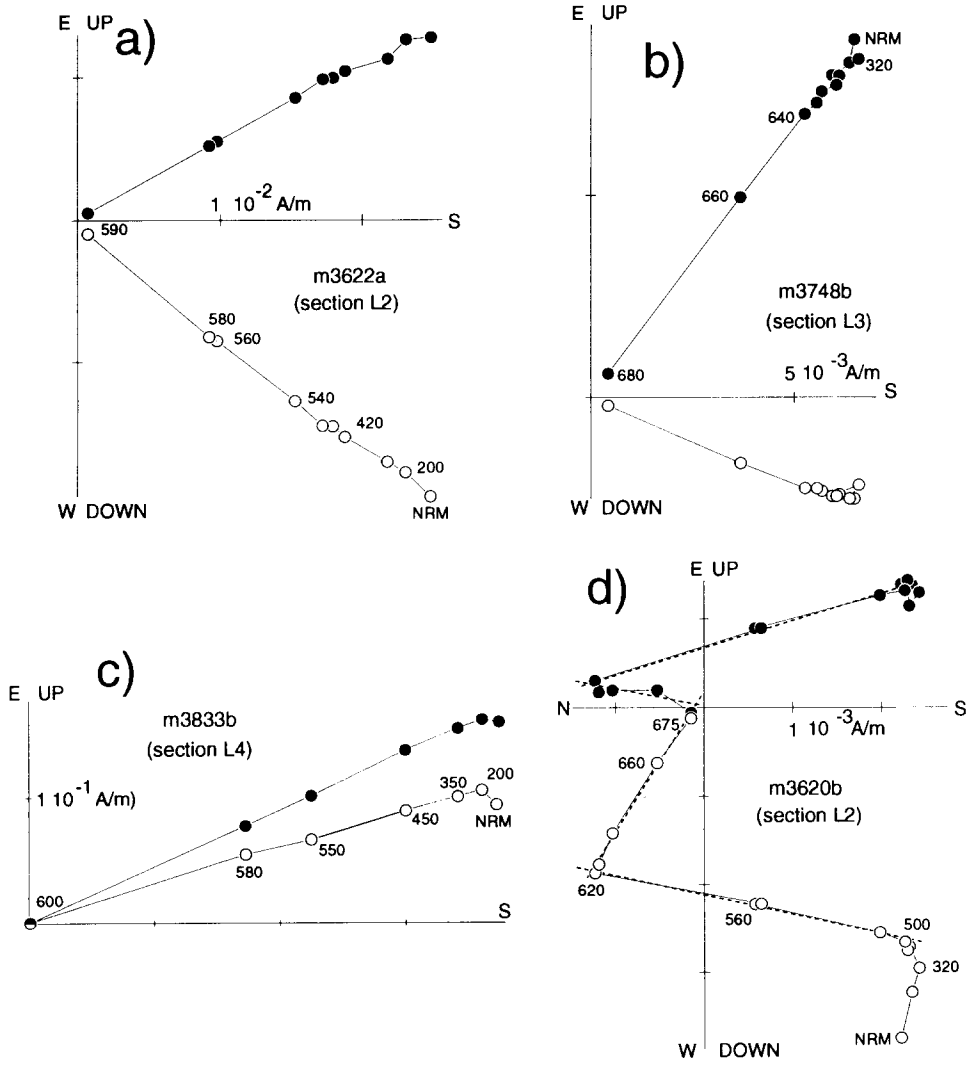


Fig. 2. Representative orthogonal plots of the Luchob Formation. (a) Section L2. (b) Section L3. (c) Section L4. (d) A weakly magnetized sample from section L2 where the high-temperature component is directed along the present-day field (*in-situ* direction is $D = 10°$, $I = 57°$), and the intermediate component is presumed to be Permian (tilt-corrected direction is $D = 155°$, $I = -58°$). All projections are *in-situ* directions. Solid symbols are projections onto the horizontal plane, open symbols onto the vertical plane. In (d) the dashed lines show the adjustment of the linear segments. Temperature is in degrees Celsius.

in this formation, mainly on lithological grounds [11]. Sandstones, siltstones and mudstones ranging in colour from brown to light grey are most common, with redbeds (the T2 member) present in the middle part of the section. This formation lies on the Paleozoic strata and is overlain by Lower–Middle Jurassic rocks with a major angular unconformity. Abundant paleoflora was found here and the formation age appears to be reliably established as Middle–Late Triassic, most probably Ladinian–Carnian and certainly pre-Rhaetian [11]. To date these are the only well dated rocks of this age in the Tien Shan. Due to their mottled appearance and friability the redbeds were not sampled. Forty-seven hand samples were taken from the other three members, from layers of varying attitude.

One hand sample oriented with a magnetic compass was taken at each stratigraphic level at each sampled section, the sampling points being spaced several metres apart along the sections. At each site in sections H1 and H2, six to eight cores were drilled from a sedimentary layer about of about 1 m in thickness; sun and magnetic compasses were used for the orientation of the cores. A fairly long stratigraphic interval was thus covered in each sampled section (Fig. 1b and d).

3. Treatments

Each hand sample was cut into two to four cubic specimens in the field using a portable diamond saw. In the laboratory the cores were cut into standard specimens. The laboratory study was started in the paleomagnetic laboratory in Moscow and was completed in Rennes. Thermal demagnetization of pilot cores was carried out in a home-made oven with an internal residual field of less than 10 nT and measurements were made with a Czechoslovakian JR-4 spinner magnetometer. In Rennes, the collection was cleaned in Schonstedt thermal demagnetizer, and Schonstedt spinner and cryogenic magnetometers were used for measurements. The results were analyzed, as usual, on stereonets and orthogonal diagrams.

The statistical treatments were somewhat different for the two types of sampling: Two specimens from a hand sample were demagnetized and a sample-mean for each identified component was computed and used for analysis. One specimen from each core was treated and all unit vectors were averaged to obtain the site-mean. Both sample- and site-means were given the same weight during analysis.

TABLE 1

Paleomagnetic results for the Luchob Formation

S	N	A/d	IN SITU				TILT CORRECTED			
			D	I	k	α_{95}	D	I	k	α_{95}
L2	9/8	173/171	153	22	18	11.6	146	-44	-	-
L3	19/18	140/36	135	-3	37	5.4	133	-39	-	-
L4	13/10	90/30	149	-20	62	5.9	163	-33	-	-
Mean (36 samples)			142	-2	14	6.2	**145**	**-39**	**23**	**4.9**
$F_{(72,72)} = 1.5$						$k_{TC}/k_{IS} = 1.6$				
Mean (3 section-means)			145	0	12	36.6	148	-39	38	20.2
$F_{(4,4)} = 6.4$						$k_{TC}/k_{IS} = 2.9$				

S = section as in the text and Fig. 1b; A/d = mean azimuth of dip/angle of dip; N = number of samples studied/accepted; D = declination; I = inclination; k = precision parameter; α_{95} = radius of 95% confidence circle; F = 95% critical value for the corresponding number of degrees of freedom in parentheses. The result printed in bold was used for interpretation.

The fold test was performed using the ratio of *in-situ* and tilt-corrected precision parameters (*PP* test) [12] and as a test (*M* test) based on the comparison of mean vectors [13]. For the latter, the distribution of bedding poles for all sampling points was divided into a number of groups irrespective of the sampling point position within a locality.

4. Results

4.1 Luchob Formation

After removal of a weak unstable magnetization, a characteristic component (ChRM) was isolated from the Luchob volcanic rocks above 300°C (Fig. 2). In just three weakly magnetized samples from section L2 was another component recognized in the final heating steps (Fig. 2d). Its *in-situ* direction is close to the present-day field (PDF). We attribute this to recent weathering and did not use this component in the analysis.

The ChRM carriers vary: hematite prevails in one section, whereas only magnetite is found in the other (Fig. 1b and c). Different horizons of the formation were probably sampled. The ChRM directions are rather well clustered in collections from three sections (Table 1). However, in ignimbrites from section L1, the ChRM directions for two specimens from a sample usually differed, by less than 10°, and the sample-means were chaotic. This was not caused by lightning strikes because the samples were taken in a gorge. We believe that large blocks of already solid lava moved with respect to each other during lava emplacement, as is often the case for such rock types. However, whatever the origin was of the scatter, this section was discarded from further consideration.

After tilt correction, three section-means are close to each other, despite the imprecise bedding (Fig. 3 and Table 1). The *PP* test is positive at the 95% confidence level, pointing to a pre-folding age for the reversed component (Table 1). The more rigorous *M* test was not used here because it is rather sensitive to small errors in structural data. Magnetization in the volcanic rocks also pre-dates deposition of the Hanaka Formation, as is shown by a positive conglomerate test on the lava pebbles (see below). The

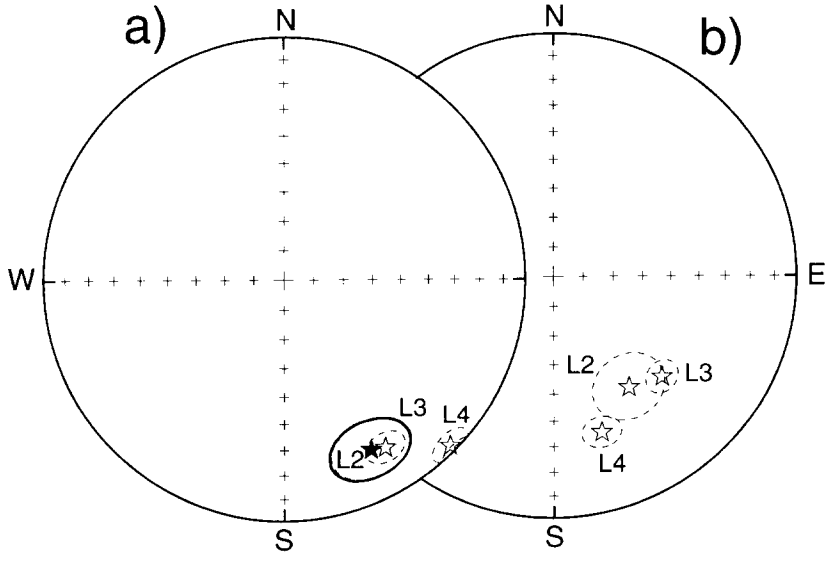


Fig. 3. Mean directions of reversed component in Luchob volcanic rocks *in-situ* (a) and after tilt correction (b). Solid symbols and solid lines are projected onto the lower hemisphere, open symbols and dashed lines onto the upper hemisphere. Mean directions are labeled as in the text and Table 1.

reversed polarity of the volcanic rocks agrees with their Early Permian age. Thus the ChRM here is most probably primary.

4.2 Hanaka ormation

The sandstones and gritstones are heavily overprinted by the PDF. This overprint often accounts for a large part of the NRM (Fig. 4a) and was destroyed at 200–250°C. After its removal, two components with negative inclinations were isolated. The first component, with a westerly declination, was usually found at intermediate temperatures, but in more weathered rocks (e.g., section H5) it was the only component present (Fig. 4b). The second component points southward and persists up to 680°C. Judging by the blocking temperatures this component resides in hematite; it shows univectorial decay to the origin (Fig. 4c). The directions of this component

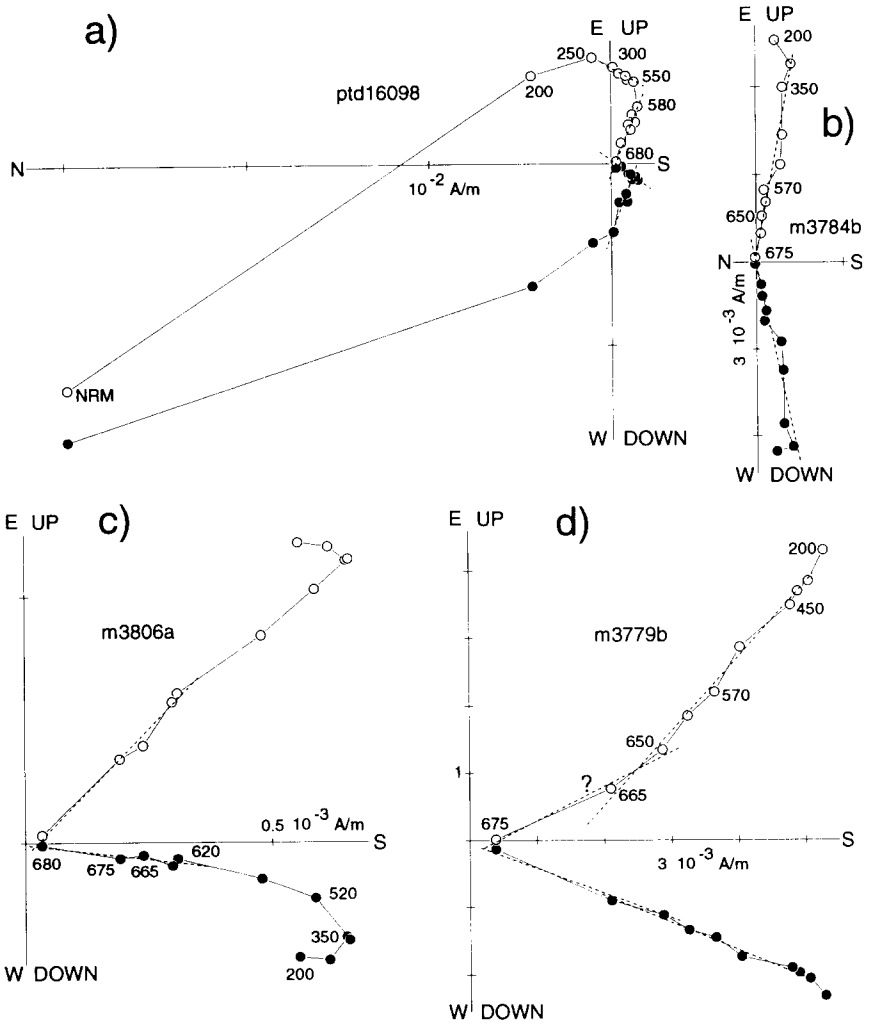


Fig. 4. Representative orthogonal plots for the main part of the Hanaka Formation (a–c) and for its upper member (d): (a) is for a sample of gritstone where both west- and south-pointing components were isolated, (b) is a sandstone with only the west-pointing component, (c) is a sandstone with only the south-pointing component, (d) is a siltstone with the south-pointing component not decaying to the origin. All projections are tilt-corrected directions. Dashed lines show the adjustment of linear segments (question mark is for a high-temperature component which was suspected in the siltstones). For clarity, NRM data were omitted from (b) and (c). Other notation as in Fig. 2.

are well grouped and their mean direction is well defined (Table 2). It is worth noting that this component was recovered both from gritstones containing debris constituted by grains of several millimetres in diameter (with almost no sandy matrix) and from fine-grained sandstones.

A distinct correlation was found between paleomagnetic directions and the characteristics of the studied outcrops. Samples taken close to the bottom of deep gorges and ravines (H1, H2, H6, Fig. 1c), often have west-pointing vectors at intermediate temperatures, and the westerly component is poorly defined (Fig. 4a) or absent at all (Fig. 4c). In contrast, the rocks sampled from small exposures in areas covered by vegetation (H5) or close to the soil (upper parts of H2 and H6) often yield a well-defined westerly component, whereas the south-pointing component might be totally absent (Fig. 4b).

On average, bedding attitudes are similar at sections H1 to H6, but some variation is present within each section. Thus, all the bedding data were pooled and their joint distribution was divided into four non-overlapping groups irrespec-

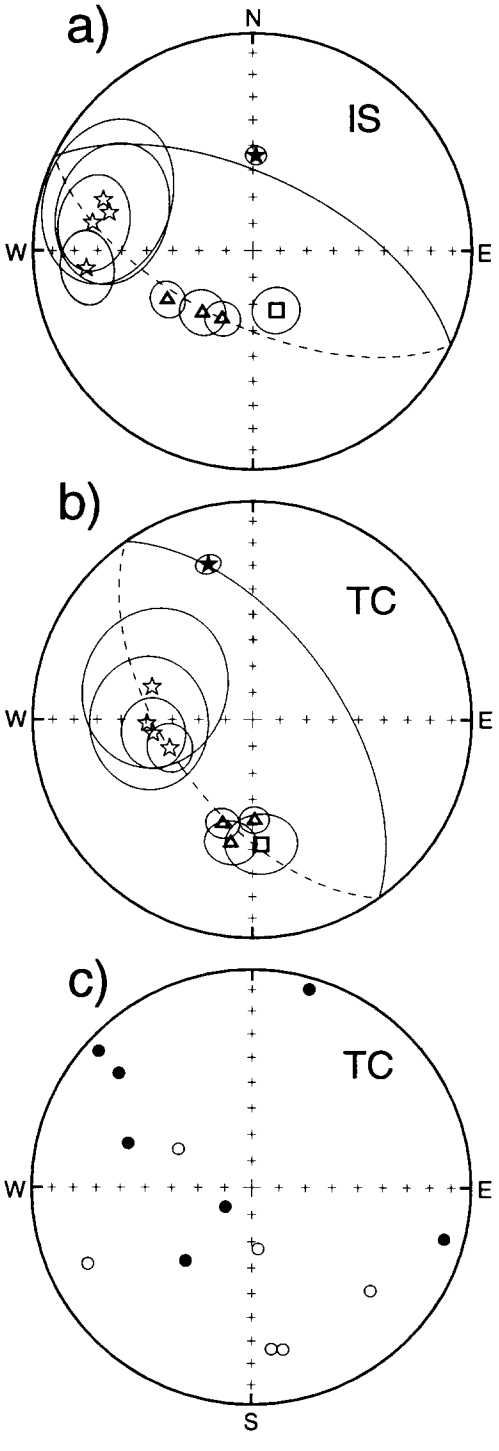
TABLE 2

Directions of south-pointing component from the Hanaka Formation

S	N	A/d	IN SITU				TILT CORRECTED			
			D	I	k	α_{95}	D	I	k	α_{95}
					The main part of the Formation					
H1	5/5		251	-49	88	6.7	197	-56	97	6.4
H2	7/5		226	-56	81	7.0	195	-41	114	5.9
H3	6/2		232	-57	-	-	203	-38	-	-
H4	7/5		219	-55	45	9.4	190	-44	44	9.4
H5	15/11		201	-65	65	5.2	177	-50	57	5.6
Mean										
	40/28		223	-59	29	4.9	**188**	**-48**	**40**	**4.2**
$F_{(54,54)}$ = 1.5 (1.4)						k_{TC}/k_{IS} = 1.4				
Groups										
I	10	316/34	240	-53	45	6.6	196	-49	57	5.9
II	8	333/28	219	-60	34	8.5	190	-42	31	8.9
III	10	314/19	204	-62	47	6.5	179	-52	72	5.2
$F_{(4,54)}$ = 2.6				f = 6.8				f = 4.0		
					The upper member of the formation					
H8	10/7	359/20	146	-58	46	7.8	157	-41	46	7.9
H9	5/4	32/37	204	-74	137	6.0	209	-38	38	11.4
Mean										
	15/11		159	-66	24	8.7	176	-42	12	12.1
$F_{(20,20)}$ = 2.1						k_{IS}/k_{TC} = 1.95				
Overall Mean										
	55/39		209	-64	17	5.5	185	-47	26	4.6
$F_{(6,70)}$ = 2.2				f = 14.3				f = 2.5		

Groups selected as in the text are labeled with Roman numerals; f = calculated value of F statistics (the critical value at the 90% confidence level is given in parentheses. Other notation as in Table 1.

tive of the sampling point position within a locality; after this, paleomagnetic mean directions for each component and each group were computed.



One of the groups proved to coincide with section H5 where only the westerly component was isolated. Due to the large scatter of the westerly component, the fold test was inconclusive. For the south-pointing component, the tilt-corrected precision parameter is larger than the *in-situ* value, but the *PP* test for the ChRM is inconclusive at the 95% level and marginal at the 90% level. At the same time, the differences between mean vectors are a little too large to pass the *M* test (Table 2), in spite of the large overlaps of the confidence circles (Fig. 5b). The best data grouping, achieved at 80% unfolding, exceeds the tilt-corrected value by just 3%. We therefore tentatively assumed a pre-folding age for the south-pointing component in these rocks. It is important to stress that all group-means are restricted to a narrow band stretching from the direction of the present field viscous component on the lower hemisphere via the westerly directions to southward-pointing vectors on the upper hemisphere (Fig. 5a and b).

In pebbles from the Hanaka conglomerates, magnetite is the NRM carrier in some samples and hematite in others, in accord with the data from the Luchob volcanic rocks. All samples revealed a unicomponent magnetization from 200°C onwards. The directions agree for two specimens from a pebble, but pebble-means are random (Fig. 5c): the calculated value of the normalized vector-resultant is 0.17 for thirteen entries while the 95% critical value of the uniformity test is 0.44 [14]. Thus the conglomerate test is certainly positive.

We propose the following model to account for the data presented above: The first component is recent and aligned with the PDF. It consists of two parts, the larger and less stable part being viscous in origin whereas the more stable

Fig. 5. Mean directions together with their confidence circles from the Hanaka Formation *in-situ* (a) and after tilt correction (b), and ChRM direction in pebbles from Hanaka conglomerates (c). ★ = low-temperature component; ☆ = westerly directions (see text for explanation); △ = directions of high-temperature component from the main part of the formation; □ = mean direction of the overprint from the upper member of the formation. Solid symbols and solid lines are projected onto the lower hemisphere, open symbols and dashed lines onto the upper hemisphere.

part, which resides in hematite, is due to chemical alteration of rocks during weathering. The presence of well-clustered paleomagnetic directions in the gritstones points to the secondary nature of this component. On the other hand, the positive conglomerate test shows that the formation was not affected by a process (e.g., a thermal event) that might have led to complete remagnetization. The acquisition of this component was probably connected with hematite formation during diagenesis, which affected the permeable sedimentary rocks but not the dense lava debris. The westerly vectors are due to the overlapping of unblocking spectra of hematite particles, which carry the recent and southward-pointing components in the sandstones and gritstones. This interpretation is confirmed by the generally curved orthogonal plots (Fig. 4a), the general data distribution (Fig. 5a and b), and the direction-outcrop correlation.

A much weaker PDF overprint was completely removed from the fine-grained sandstones and siltstones of the upper formation member below 300°C. No trace of westward-pointing directions was found and a well-defined south-upward-pointing component was readily isolated here, but the orthogonal plots do not decay to the origin and another component could be possible. Both components appear to have similar declinations, but the high- temperature component has a gentler inclination (Fig. 4d). However, the weaker component was represented by no more than three points on the orthogonal plots of a few samples and we decided not to take this component into account. For this member, the *PP* test [12] points to a post-folding origin of the main component, but the calculated *F* statistics in any coordinate system [13] much exceed the critical value. This remanence can therefore be considered as neither pre- nor post-folding in age. At the same time, the mean directions of the main component from both parts of the formation agree well after tilt correction but diverge *in-situ* (Fig 5 and Table 2). Also interesting is that H8 and H9 mean vectors after tilt correction have similar inclinations and divergent declinations. Because the mean strikes of sections H8 and H9 also differ in the same sense as the observed paleomagnetic declinations and because these rocks were sampled from a narrow, fault-bounded strip, rotation may have occurred. We calculated the overall mean for the Hanaka Formation (Table

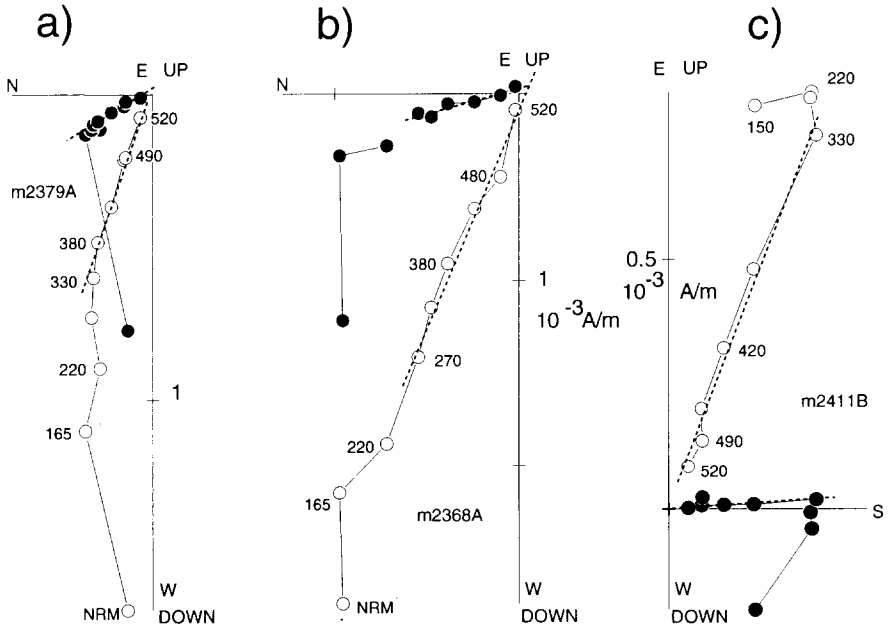


Fig. 6. Representative orthogonal plots for the T1 member (a) and T4 and T5 members (b and c) of the Madighen Formation. All projections are tilt-corrected directions. Dashed lines show the adjustment of linear segments. For clarity, the NRM direction was omitted from (c). Other notation as in Fig. 2.

2), but, taking into account the limited number of data for the upper member and their controversial character, the result from the main part of the formation will be used for tectonic interpretation.

After removal of the viscous overprint, all the measured directions and isolated components, without a single exception, have reversed polarity. In the latest Permian (mid-Tatarian time), the predominantly reversed Kiaman chron was replaced by the mixed polarity Illawara chron [15]. It is unlikely that the whole formation would have been overprinted during the Illawara or later with no trace of normal polarity. Thus this formation and all the components isolated here should be Permian in age.

4.3 Madighen Formation

NRM values are about $1-2\times10^{-3}$ A/m in these lacustrine sedimentary rockss, and magnetization intensity often drops several-fold after heating to 200°C. Many samples from the T1 member of the formation (sections M1 and M3, Fig. 1d) were unstable or showed high within-sample scatter of unit vectors, and most of the results come from the grey siltstones of the upper two members (T3 and T4, Fig. 1d). The ChRM was reliably isolated from many of the samples after removal of the low-temperature component (Fig. 6).

Normal and reversed ChRM directions from members T3 and T4 are antipodal (Fig. 7), and

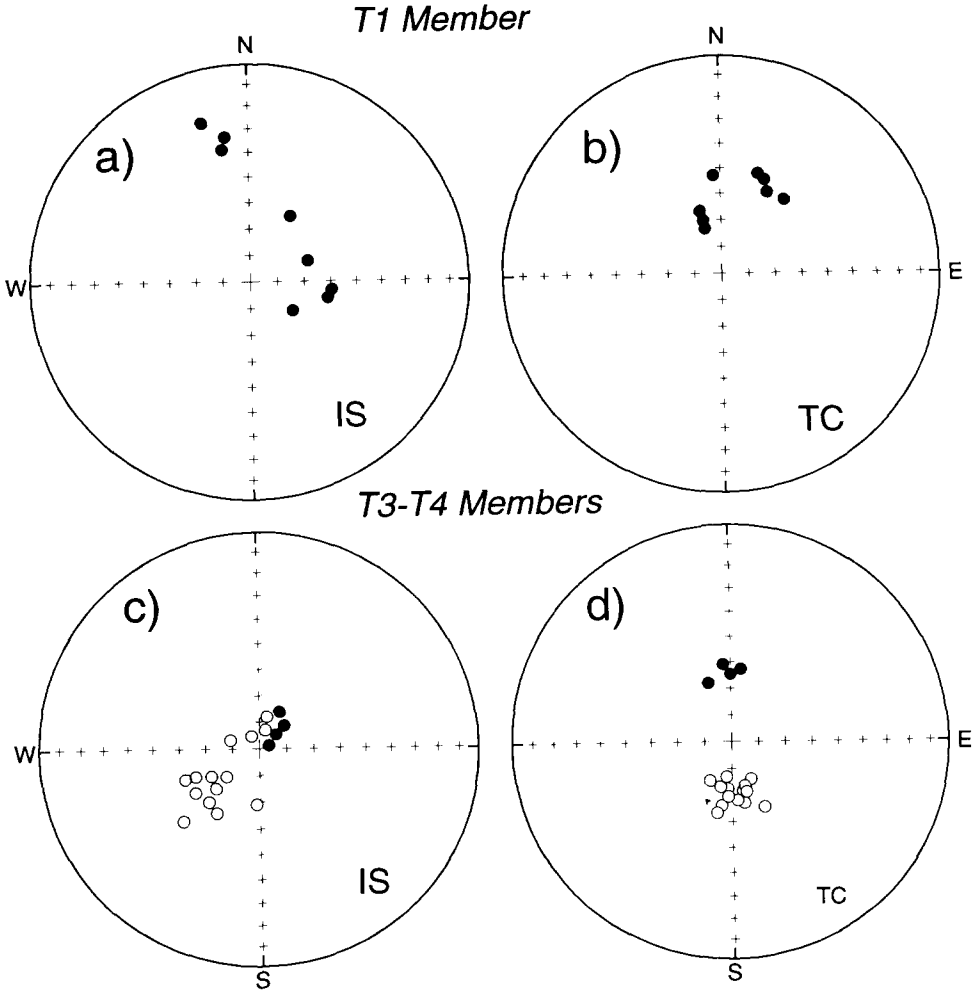


Fig. 7. ChRM directions from T1 (a and b) and T3 and T4 (c and d) members of the Madighen Formation *in-situ* (a and c) and after tilt correction (b and d). Solid symbols are projected onto the lower hemisphere, open symbols onto the upper hemisphere.

both modifications of the fold test point to a pre-folding age (Table 3). These Middle–Upper Triassic rocks are overlain, with a major angular unconformity, by Lower Jurassic strata. Thus the bipolar ChRM in the upper part of the Madighen Formation is almost certainly primary. The *PP* test is positive for the T1 data, whereas the *M* test is not (Table 3). The latter result may be due to the limited size of the collection. Nevertheless, we decided not to combine two datasets and the T1 data were not used.

5. Interpretation

It seems logical to compare results from Central Asia with the apparent polar wander path (APWP) for Siberia [16]. However, the data used for the Permian–Cenozoic part of the Siberian APWP are not based on complete demagnetization and component analysis and are thus unreliable. The European APWP, which is based on better quality data and confirmed by the North America APWP [17], will be used in our interpretation. The reference inclinations and declinations were placed in the centres of the corresponding time windows used for the APWP construction [17].

Seeing as the Early Permian lasted from 295 Ma to 258 Ma [18], an age of 276 ± 19 Ma was assigned to the data from the Lower Permian Luchob Formation. The termination of the Kiaman reversed superchron is usually placed in the mid-Tatarian (ca. 247 Ma), and although there are indications that it may be older [19] we prefer to use the more common dating. The Hanaka rocks are younger than the Luchob volcanic rocks although the former acquired their magnetization at the same time as the latter—during the Kiaman reversed superchron. We arbitrarily place the Hanaka results at the Early–Late Permian boundary (258 Ma) and assigned to it an age error stretching to the termination of the Kiaman superchron. The Madighen Formation was reported to be probably Ladinian–Carnian and cer-

TABLE 3

Paleomagnetic results for the Madighen Formation

S	N	A/d	IN SITU				TILT CORRECTED			
			D	I	k	α_{95}	D	I	k	α_{95}
					T1 member					
M2	5/5	351/32	84	67	28	12.0	30	52	37	10.3
M4	4/3	176/42	348	32	120	7.4	328	72	283	4.8
Mean	9/8		27	64	6	21.0	17	62	17	12.2
$F_{(14,14)} = 2.5$						$k_{TC}/k_{IS} = 2.9$				
$F_{(2,12)} = 3.9$				f = 37.3				f = 15.8		
					T3 and T4 members					
M5,6	11/7	343/23	74	87	60	6.8	349	66	76	6.8
M7,8	9/6	259/27	47	55	31	10.3	1	72	52	8.0
M9	8/5	281/25	56	62	40	9.9	0	71	84	6.8
Mean	28/18		52	70	18	7.7	**355**	**70**	**68**	**4.0**
$F_{(4,30)} = 2.7$				f = 11.5				f = 0.9		

All data are inverted to normal polarity; notation as in Table 1.

tainly pre-Rhaetian in age [11], thus covering the interval from 235 Ma to ca. 212 Ma, and a 223 ± 12 Ma age was assigned.

In spite of the age uncertainties associated with our data, the agreement of the Luchob and Hanaka inclinations with the reference values is good (Fig. 8a). Indeed, both can be shifted within their age uncertainties, and even slightly beyond them, and still remain compatible with the reference curve. The Madighen inclination is somewhat steeper than the reference one, and would better agree with the younger reference value (205 ± 10 Ma). We do not believe that the difference is due to tectonic movements. Instead, this discrepancy is more likely to be due to some misfit in age, either in the rocks themselves or in the reference data, but addressing this problem in more detail is out of the scope of this paper.

In contrast, all the declinations are deflected westward with respect to the European data beyond any possible errors (Fig. 8b). A tectonic rotation therefore appears to be the only acceptable explanation for these westerly declinations. The south Tien Shan was affected by Alpine deformation, which was taken into account with the aid of paleomagnetic data. A collection of Cretaceous rocks was made at locality 1 and Oligocene–Lower Miocene rocks were sampled at locality 3. The Cretaceous rocks revealed a mean direction that is rotated about 20° counterclockwise with respect to the Eurasian mean declination [our unpublished results]. The Oligocene–Miocene mean vector has been rotated in the same sense, by ca. 25°; in addition, a similar late Cenozoic rotation of the entire Fergana basin and adjacent ranges has been deduced from Tertiary [1] and Cretaceous [2] paleomagnetic data. After subtraction of these younger rotations, the measured declinations are still deflected westward (Fig. 8b).

Several attempts to obtain Jurassic paleomagnetic data in the Tien Shan have failed [2,20; A.F. Eroshkin, pers. commun.], and it is impossible directly to fill the gap in the paleomagnetic record. Some characteristics of the Jurassic sedimentation are thought to reflect differential subsidence and uplift, and in turn these vertical movements have been correlated with tectonic cycles from other parts of the Alpine belt [21]. However, apart from local weak angular unconformities at

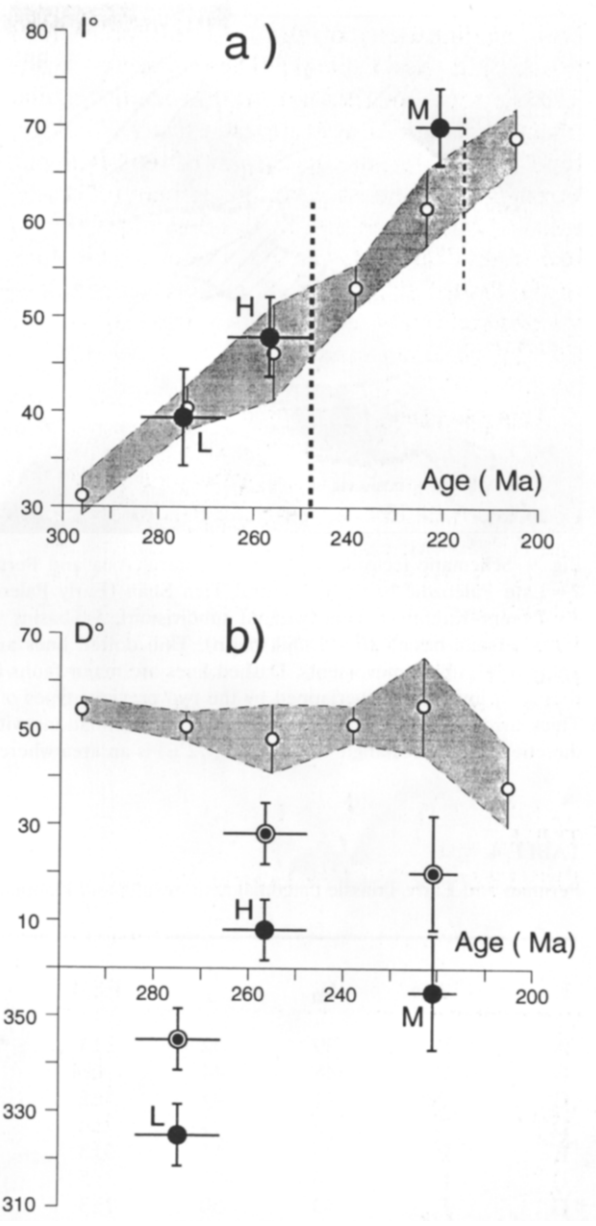


Fig. 8. Plots of inclination (a) and declination (b) versus age. ● = our data (*L*, *H* and *M* are the Luchob, Hanaka and Madighen formations); 'double circles' = declinations corrected for the Alpine rotation (see text for explanation); ○ = European reference directions [17]. Solid vertical lines are error bars for paleomagnetic data. Shaded area is the confidence band for reference directions. Solid horizontal lines are age error bars as described in the text. Thick vertical dashed line is the Kiama–Illawara boundary; thin vertical dashed line is the Norian–Rhaetian boundary.

the Jurassic–Cretaceous boundary there is little evidence of tectonism, and throughout the Jurassic up to the end of the Paleogene neither folding

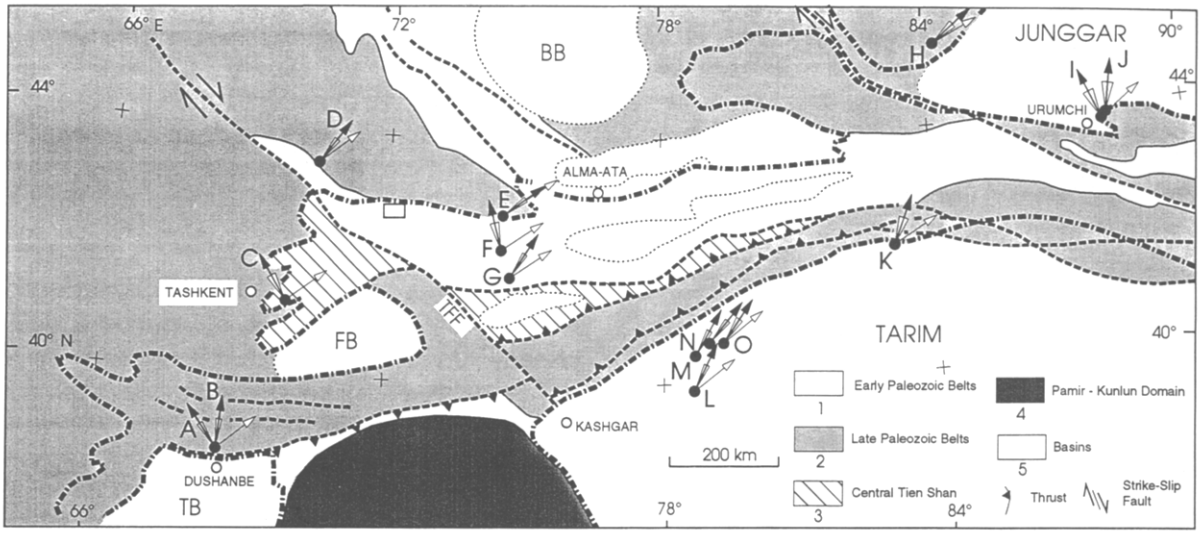


Fig. 9. Schematic tectonic zonation of Central Asia and Permian–Early Triassic paleomagnetic data. *1* = Early Paleozoic belts; *2* = Late Paleozoic belts; *3* = central Tien Shan (Early Paleozoic structures strongly reworked by Late Paleozoic movements); *4* = Pamirs–Kunlun domain (without subdivision); *5* = basins filled mainly with Mesozoic–Cenozoic rocks (*BB* = Balkhash basin; *FB* = Fergana basin; *TB* = Tadjik basin). Thin dotted lines are basin boundaries. Thick dot–dash line marks the limit of areas uplifted by Alpine movements. Dashed lines are major faults (*TFF* = Talas–Fergana fault). The boundaries of tectonic zones are thin solid lines if not overlapped by the two previous types of lines. ● = paleomagnetic sampling localities labeled as in Table 4. Thick arrows with black heads are measured declinations with confidence sectors. Thin arrows with white heads are reference directions. Small rectangle (at ca. 43°N, 72°E) is an area where counterclockwise rotations were deduced from structural data [43].

TABLE 4

Permian and Early Triassic paleomagnetic results and kinematic parameters for Central Asia

L	Age	I_m	I_r	F±ΔF	Dm	Dr	R±ΔR	REF
A	Pe	39	42	3±4	325	51	86±6	TP
B	Pl	48	44	-4±4	8	49	41±7	TP
C	Pe	41	44	3±2	327	52	85±3	[22]
D	Pe	42	49	7±4	36	52	16±5	BK
E	P	46	51	5±3	55	56	1±5	[23]
F	P	48	51	3±5	343	56	73±8	[23]
G	P	49	50	1±3	33	55	22±5	[23]
H	Pe	50	57	7±4	48	61	13±5	[24]
I	Pl	62	61	-1±4	343	57	74±8	[25]
J	Pl	59	62	3±7	7	58	51±10	[20]
K*	Pl-Te	55	61	6±4	22	55	33±9	[26]
L	Pl	50	54	4±4	28	52	24±7	[27]
M	Pe	58	51	-7±3	32	56	24±5	[28]
N	Pe	50	51	1±3	42	56	14±5	[29]
O	Pe	41	51	10±4	34	56	22±6	[29]

* The average of Late Permian and Early Triassic reference values was used. *L* = locality as labeled in Fig. 9; *D* and *I* = declination and inclination (inverted to normal polarity); *F* = flattening with interval of confidence; *R* = rotation angle (positive values stand for anticlockwise rotation) with interval of confidence. Subscripts 'm' and 'r' denote measured and reference data respectively. The confidence limits were computed as suggested in [32]. References: *TP* = this paper; *BK* = Bazhenov and Klishevich, unpublished data.

nor thrusting occurred in the western Tien Shan. A Jurassic age for the rotations is therefore unlikely, because rotations of several tens of degrees should have somehow been reflected in the geological record. We do note, however, a large unconformity between the Middle–Late Triassic Madighen Formation and Jurassic rocks. Thus, a Permian–Triassic age for the rotations appears to be the most likely solution. Moreover, the significant difference between the Early Permian and younger results implies that the rotation was not an instantaneous movement but was distributed in time.

Unfortunately, it is difficult to extrapolate the above conclusion to a larger area. As noted earlier, Upper Permian rocks are rare and Middle–Upper Triassic (pre-Rhaetian) rocks are absent in the western Tien Shan, and we know of no way to check our findings against other areas. The necessary data may come from western China, but Triassic rocks are scarce there too.

6. Regional analysis

Permian and Lower Triassic results are known from the Chatkal range [22], north Kirghizia [23], the western [24] and southern [20,25] periphery of the Junggar basin, and north Tarim [26–29] (Fig. 9 and Table 4). In addition, a result from Permian redbeds in south Kazakhstan [Bazhenov and Klishevich, unpublished] was used. The result from the Heavenly Lake section [23] is based on seven sites, and one of these is clearly an outlier. Its omission led to a small shift in the mean vector and a five-fold increase in the precision parameter (24 compared to 129); we preferred the modified result (Table 4). Permian data from around the Fergana basin [e.g., 6] are not based on complete demagnetization and component analysis and their reliability is unclear. We decided to use only completely demagnetized data.

Irrespective of the age of the studied rocks, Permian and Early Triassic inclinations show good agreement with the reference values, the differences being either insignificant or slightly significant but not systematic (Table 4). Thus the extended dataset confirms the conclusion [16,23,24, 30,31] that Tarim, Tien Shan, Kazakhstan and the European plate were already welded together by the mid-Permian.

In contrast, all the declinations (with the sole exception of result E) are deflected westward with respect to the reference data (Fig. 9 and Table 4). These deflections are too large and systematic to be accounted for by errors. Although the angles of rotation differ, their sense is the same throughout the area. Declination anomalies were found in the Early Paleozoic (localities F and G; Fig. 9 and Table 4) and Late Paleozoic foldbelts (A–C), in northern Tarim (K–O), and even in the areas where Permian rocks were deformed only by Alpine movements (I and J). Thus we may assume that the rotation is regional in scale. Unfortunately, little is known of the size and limits of the rotated area. The unrotated Permian direction was reported from a locality close to the Tien Shan–Kazakhstan border [23]. Minimal rotation was found in the northern part of Junggar [24] and south Kazakhstan.

These movements are certainly younger than Early Permian and are probably post-Late Permian, judging by the age of the formations studied. The southwestern part of the Tien Shan was rotated counterclockwise in the Cenozoic [1,2], but a considerable part of its rotation is older (Fig. 8b). To the north and east of the Talas–Fergana fault, paleomagnetic data on the post-Triassic rocks of Kirghizia [1], the south Junggar basin [33] and northern Tarim [34] are statistically identical to the reference data, even at localities close to those which revealed rotated Permian directions (e.g., in northern Tarim). The possibility of Jurassic rotation has already been disproved in the previous section. Thus the observed declination anomalies are Late Permian–Triassic in age, which is compatible with the existing views on Central Asia tectonics. The importance of Triassic deformation has been well recognized in the Pamirs [35] and Afghanistan [36] and advocated for Mangyshlak and the western part of the Turan plate [37]. In the area of the present study, the apparent absence of such movements is most probably due to the gap in geological record.

Paleomagnetic data from Tarim have been used for construction of the APWP, which has been compared with the Siberian data. The observed difference between these two APWPs was accounted for by a post-Permian movement of the Tarim block as a rigid body to the northeast [38]. This approach, which is well-known for large

plates, is hardly valid here, because, in addition to the low reliability of the Siberian data, there are a number of serious objections (e.g., the calculation of the kinematic parameters is valid only for a rigid plate, the latter being doubtful as far as Tarim is concerned). The most important constraint, however, is that of the geology of Central Asia. According to geological and paleomagnetic data, the closure of the Junggar ocean [39] and the collision of Tarim with the Tien Shan occurred before the end of the Permian [40]. The main tectonic zones of the Tien Shan are easily recognizable outside the Junggar area (Fig. 9), the Early Paleozoic foldbelt being traceable well into the southern part of Kazakhstan and the Late Paleozoic structures of the south Tien Shan extending to the Aral Sea [7]. These zones are identified on the basis of Early and middle Paleozoic geology, and younger rotations should not have disrupted the now continuous zonation. In particular, the proposed northeast movement of the Tarim block [38] implies that it should have crossed the area which had already been occupied by the southern Tien Shan. Finally, the close agreement of the measured inclinations and the Eurasian reference data should be taken into account. We tried to move as a rigid block the entire area where counterclockwise rotations were found and failed to find a Euler pole position for which these two data sets would remain compatible. There is therefore neither the space nor structure to accommodate large post-Permian displacement of a Tarim-size block in Central Asia. Of course, older structures could have been —and most probably were—disrupted and deformed by younger movements. For instance, the Talas–Fergana fault clearly affects the tectonic pattern in the central part of the Tien Shan realm, but the 200 km offset along this fault [41] is rather small compared to the size of the area as a whole (Fig. 9).

Thus, declination anomalies have probably resulted from movements of small blocks. Rotations might have been mutually independent, and the declination pattern may just be a coincidence. But this is not a convincing argument. Much more attractive is the assumption that a unique mechanism is responsible, and domino-fashion rotations under a strike-slip regime appear to be the most likely. Of course, more paleomagnetic data are needed to place additional constraints on the spatial and temporal distribution of such rotations. Perhaps more necessary is a detailed structural study of the entire belt. However, owing to large angles of rotation that are capable of creating a complicated fault network [42], strong Alpine deformation of the belt which could, and probably did, rework the older pattern, and the scarcity of Late Paleozoic and Triassic formations this will not be an easy task. So far, only a few counterclockwise rotations of unknown age have been found in the western part of the Kirghiz range [43], where the fault network is similar to that in a sinistral strike-slip zone. In general, we know no examples of strike-slip movements that have been successfully and reliably proven over large areas which were then later reworked by younger movements.

The above hypothesis implies neither uniformity nor strict synchroneity of strike-slip movements in Central Asia. Also, it may be that the rotation deduced from the Middle–Late Triassic result in the west Tien Shan affected a much larger area. It is interesting to note that the onset of terrigenous sedimentation occurred in Rhaetian–Early Jurassic time. On the basis of some geological data these coal-bearing sequences are limited to small basins. Could these basins be of the pull-apart type, related to the final stage of strike-slip deformation?

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References

1 J.C. Thomas, H. Perroud, P.R. Cobbold, M.L. Bazhenov, V.S. Burtman, A. Chauvin and E. Sadybokasov, A paleomagnetic study of Tertiary formations from the Kirghiz Tien Shan and its tectonic implications, J. Geophys. Res., in press, 1993.

2 M.L. Bazhenov, Cretaceous paleomagnetism of the Fergana basin and adjacent ranges: tectonic implications, Tectonophysics, in press, 1992.

3 P. Molnar and P. Tapponnier, Cenozoic tectonics of Asia: effects of a continental collision, Science 189, 419–426, 1975.
4 A.L. Didenko and D.M. Pechersky, Paleomagnetism of the Middle Paleozoic ophiolite complexes of the Alay Range, Geotektonika 4, 56–68, 1988 (in Russian).
5 Yu.S. Rzhevsky, Paleomagnetism of magmatic and sedimentary rocks of the Ghissar Range (South Tien Shan), in: Magnetostratigraphy and Paleomagnetism of Sedimentary and Volcanic Units of the USSR, A.N. Khramov, ed., pp. 64–78, VNIGRI, Leningrad, 1986 (in Russian).
6 K.P. Panov, Paleomagnetic studies of Variscan structures, South-Western Fergana (abstract), IV All-Union Geomagnetic Symp. Proc., Academia Nauk, Vladimir, 2, pp. 152–153, 1991 (in Russian).
7 V.E. Khain, Regional Geotectonics (Extra-Alpine Europe and Western Asia), 360 pp., Nedra, Moscow, 1977 (in Russian).
8 A.S. Masumov, O.M. Borisov and F.R. Bensh, Late Paleozoic of the Central and Southern Tien Shan, 176 pp., FAN, Tashkent, 1978 (in Russian).
9 R.B. Baratov, E.N. Goretskaya and S.N. Shchukin, Dacite-liparite formation of the Southern Ghissar, 140 pp., Donish, Dushanbe, 1973 (in Russian).
10 V.S. Luchnikov, The Hanaka Formation of the Ghissar Range: Problems of oil-prospecting in Tadjikistan, Donish, Dushanbe 6, 204–207, 1974 (in Russian).
11 I.A. Dobruskina, Stratigraphic Position of Plant-bearing Triassic Rocks from Eurasia, 163 pp., Nauka, Moscow, 1976 (in Russian).
12 M.W. McElhinny, Statistical significance of the fold test in palaeomagnetism, Geophys. J. R. Astron. Soc. 8, 338–340, 1964.
13 P.L. McFadden and D.L. Jones, The fold test in palaeomagnetism, Geophys. J. R. Astron. Soc. 67, 53–58, 1981.
14 K.V. Mardia, Statistics of Directional Data, 357 pp., Academic Press, London, 1972.
15 A.N. Khramov, G.I. Goncharov, R.A. Komissarova, S.A. Pisarevsky, I.A. Pogarskaya, Yu.S. Rzhevsky, V.P. Rodionov and I.P. Slautsitais, Paleomagnetology, 312 pp., Nedra, Leningrad, 1982 (in Russian).
16 A.N. Khramov, E.L. Gurevich, R.A. Komissarova, R.A. Osipova, S.A. Pisarevsky, V.P. Rodionov and I.P. Slautsitais, Paleomagnetism, microplates and Siberian plate consolidation, J. Geodyn. 2, 127–139, 1985.
17 R. Van der Voo, Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions, Rev. Geophys. 28, 167–206, 1990.
18 G.S. Odin and C. Odin, Echelle numerique des temps geologiques, Geochronique 35, 12–21, 1990.
19 M. Haag and F. Heller, Late Permian to Early Triassic magnetostratigraphy, Earth Planet. Sci. Lett. 107, 42–54, 1991.
20 S. Nie, R. Van der Voo, M. Li and D.B. Rowley, Paleomagnetism of Late Paleozoic rocks in the Tianshan, Northwestern China, Tectonics, in press, 1992.
21 I.G. Tscherba, The reflection of the Alpine tectonic phases in Mesozoic–Cenozoic sequences of the Tien shan, Geotektonika 2, 42–53, 1990 (in Russian).
22 A.S. Bolshakov, G.M. Solodovnikov and Yu.K. Vinogradov, Paleointensity of Early Permian geomagnetic field, Izv. Akad. Nauk SSSR, Ser. Fiz. Zemli 7, 70–77, 1989 (in Russian).
23 M. Audibert and M.L. Bazhenov, Permian paleomagnetism of the North Tien Shan: tectonic implications, Tectonics 11, 1057–1070, 1992.
24 X. Zhao, R.S. Coe, Y. Zhou, H. Wu and J. Wong, New paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan, Tectonophysics 181, 43–81, 1990.
25 R. Sharps, Y. Li and M. McWilliams, Paleomagnetic investigation of Upper Permian sediments in the South Junggar Basin, China, J. Geophys. Res. 92, 1753–1765, 1992.
26 P.L. McFadden, X. Ma, M.W. McElhinny and Z. Zhang, Permo-Triassic magnetostratigraphy in China: northern Tarim, Earth Planet. Sci. Lett. 87, 152–160, 1988.
27 Y.P. Li, M. McWilliams, A. Cox, R. Sharps, Y.A. Li, Z. Gao, Z. Zhang and Y. Zhai, Late Permian paleomagnetic pole from dykes of the Tarim Craton, Geology 16, 275–278, 1988.
28 R. Sharps, M. McWilliams, Y. Li, A. Cox, Z. Zhang, Y. Zhai, Z. Gao, Y.A. Li and Q. Li, Lower Permian paleomagnetism of the Tarim block, northwestern China, Earth Planet. Sci. Lett. 92, 275–291, 1989.
29 Y. Bai, G. Chen, Q. Sun, Y. Sun, Y. Li, Y. Dong and D. Sun, Late Paleozoic polar wander path for the Tarim platform and its tectonic significance, Tectonophysics 139, 145–153, 1987.
30 B.F. Windley, M.B. Allen, C. Zhang, Z.Y. Zhao and G.R. Wang, Paleozoic accretion and Cenozoic redeformation of the Chinese Tien Shan range, central Asia, Geology 18, 128–131, 1990.
31 L.P. Zonenshain, M.I. Kuzmin and L.M. Natapov, in: Geology of the USSR: A Plate Tectonic Synthesis, B.M. Page, ed., AGU Geodyn. Ser. 21, 242 pp., 1990.
32 H.H Demarest, Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, J. Geophys. Res. 88, 4321–4328, 1983.
33 Y. Chen, J.P. Cogne, V. Courtillot, J.P. Avouac, P. Tapponnier, G. Wang, M. Bai, H. You, M. Li, C. Wei and E. Buffetaut, Paleomagnetic study of Mesozoic continental sediments along the Northern Tien Shan (China) and heterogeneous strain in Central Asia, J. Geophys. Res. 96, 4065–4082, 1991.
34 Y.P. Li, Z. Zhang, M. McWilliams, R. Sharps, Y. Zhai, Y.A. Li, Q. Li and A. Cox, Mesozoic paleomagnetic results of the Tarim craton: Tertiary relative motion between China and Siberia?, Geophys. Res. Lett. 15, 217–220, 1988.
35 S.V. Ruzhentsev, Tectonic Evolution of the Eastern Pamirs and the Role of Horizontal Movements in its Alpine Tectonics, 202 pp., Nauka, Moscow, 1968 (in Russian).
36 A.M.C. Sengor, The Cimmeride Orogenic System and the Tectonics of Eurasia, Spec. Pap. Geol. Soc. Am. 195, 82 pp., 1984.
37 V.I. Popkov, Tectonics of pre-Jurassic sedimentary complex of the western Turan plate, Geotektonika 4, 106–116, 1986 (in Russian).
38 Y.P. Li, An apparent polar wander path from the Tarim block, China, Tectonophysics 181, 31–41, 1990.

39 A.R. Carroll, Y. Liang, S.A. Graham, X. Xiao, M.S. Hendrix, J. Chu and C.L. McKnight, Junggar basin, northwest China: trapped Late Paleozoic ocean, Tectonophysics 181, 1–14, 1990.
40 R.G. Coleman, Continental growth of northwest China, Tectonics 8, 621–635, 1989.
41 V.S. Burtman, The Structural Evolution of Paleozoic Fold Systems, 164 pp., Nauka, Moscow, 1976 (in Russian).
42 A. Nur, H. Ron and O. Scotti, Fault mechanics and the kinematics of block rotation, Geology 14, 746–749, 1986.
43 V.V. Kiselev and V.G. Korolev, Strike-slips and Rotational Structures in the Western Part of the Kirghiz Range (Reports on Geology of the Tien Shan), pp. 147–151, Ilim, Frunze, 1964 (in Russian).